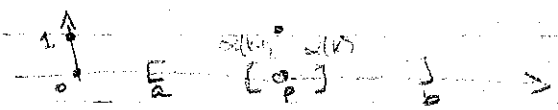
Math 104, Homework #5

1a. Take a partition $P = \{a < a_0 < \dots < a_{n-1} < b\}$ and $\Delta\alpha_i = i$

Note that $p \in [\alpha(\ell-i), \alpha(\ell)]$ for some ℓ since P partitions $[a,b]$



So, $U(f,P,\alpha) = \sum_{k=1}^{n} M(f, [\alpha(k-i), \alpha(k)]) \cdot (\alpha(k) - \alpha(k-i))$

$= 0 + \dots + 0 + f(p) \cdot (\alpha(\ell) - \alpha(\ell-i)) + 0 + \dots + 0$

(1) $= \alpha(\ell-i) - \alpha(\ell)$

$\because f(x) = 0$ for $x \neq p$, and so $f(p) = \max(f(x))$ on $[\alpha(\ell-i), \alpha(\ell)]$

Also, $L(f,P,\alpha) = \sum_{k=1}^{n} m(f, [\alpha(k-i), \alpha(k)]) \cdot (\alpha(k) - \alpha(k-i))$

(2) $= 0 + \dots + 0 = 0$

$\because f(x) = 0$ for all intervals away from $x = p$, and for the interval around $x = p$, $[\alpha(\ell-i), \alpha(\ell)]$ must contain at least one other point, say x_j, since α mon. incr. and $f(x_j) = 0$

Combining (1) and (2),

(3) $U(f,P,\alpha) - L(f,P,\alpha) = \alpha(\ell) - \alpha(\ell-i)$

Note that α is cont.s, so, $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $|\ell - (\ell-i)| = |i| < \delta$ implies $|\alpha(\ell) - \alpha(\ell-i)| = \alpha(\ell) - \alpha(\ell-i) < \varepsilon$

(don't need absolute values b/c α is mon. incr. so $\alpha(\ell) > \alpha(\ell-i)$)

(4) So for $|i| < \delta$, $U(f,P,\alpha) - L(f,P,\alpha) < \varepsilon$

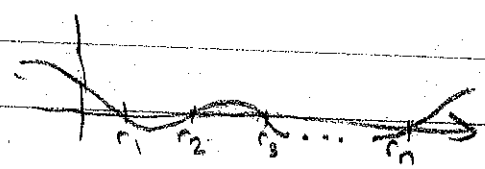
From (4), $f(x)$ is Riemann integrable on $[a,b]$

Also, from (2), $L(f,P,\alpha) = 0$, and from (3), $U(f,P,\alpha)$ can be made arbitrarily close to 0

$$\Rightarrow \int_a^b f\, d\alpha = 0 \qquad \square$$

(0) b. Let $f(x) = a_n x^n + \dots + a_1 x + a_0$
s.t. $a_n \neq 0$, so $\deg(f(x)) = n$

$f(x)$ has n distinct roots in $\mathbb{R}$, say $r_1, \dots, r_n$



To show: $f'(x)$ has at least $(n-1)$ roots

For each $i = 2, \dots, n$, $\exists c_i \in (r_{i-1}, r_i)$ s.t.
$$f'(c_i) = \frac{f(r_i) - f(r_{i-1})}{i - (i-1)} = 0 \quad \text{by MVT}$$

So, $f'(x) = 0$ at least $(n-1)$ times for each of the
(1) $n-1$ intervals defined by ?
Note that since the intervals are distinct and there is one r_i per interval, the r_i are distinct as well.

0

⌈To show: $f'(x)$ has at most $(n-1)$ roots⌋

Recall $f(x) = a_n x^n + \dots + a_1 x + a_0$ from (0)

So, by power rule and properties of differentiation,

$$f'(x) = a_n \frac{d}{dx}(x^n) + \dots + a_1 \frac{d}{dx}(x) + a_0 \frac{d}{dx}(1)$$
$$= a_n \cdot n\, x^{n-1} + \dots + a_1 (1) + 0$$

So, $f'(x)$ is of degree $(n-1)$ (since $a_n \neq 0$)

(1)

Set $f'(x) = 0$:

(2) $$a_n \cdot n \cdot x^{n-1} + \dots + a_2 \cdot 2 \cdot x + a_1 = 0$$

From abstract algebra, (2) can be factored as:

$$(x + q_1) \dots (x + q_{n-1}) = 0$$

Note that we divided out the first coefficient, $a_n \cdot n$, and that $q_1, \dots, q_{n-1}$ are not necessarily real

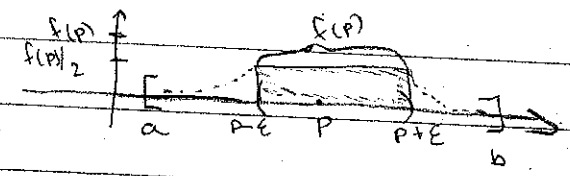
(3) $\Rightarrow f'(x)$ has at most $(n-1)$ roots, $q_1, \dots, q_{n-1}$

Combining (1) and (3), $f'(x)$ has exactly $(n-1)$ distinct roots $\square$

c. (proof by contradiction)

Suppose that $f(p) \neq 0$ for some $p \in [a,b]$

f cont.s $\Rightarrow$ there is a ball around $x=p$, say $(p-\varepsilon, p+\varepsilon)$ where $|f(x)| > f(p)/2$ (we can remove absolute values since $f(x) \geq 0 \quad \forall x \in \mathbb{R}$)



Then $$\int_a^b f(x)\,dx = \underbrace{\int_a^{p-\varepsilon} f(x)\,dx}_{A} + \underbrace{\int_{p-\varepsilon}^{p+\varepsilon} f(x)\,dx}_{B} + \underbrace{\int_{p-\varepsilon}^{b} f(x)\,dx}_{C}$$

Note that for $x \in [a, p-\varepsilon] \cup [p-\varepsilon, b]$, $f(x) \geq 0$ so $U(f,\alpha) \geq 0$ and $L(f,\alpha) \geq 0$

(1) for any $\alpha(x) \Rightarrow A, C \geq 0$

Also, $$B := \int_{p-\varepsilon}^{p+\varepsilon} f(x)\,dx \geq \int_{p-\varepsilon}^{p+\varepsilon} f(p)/2\,dx = \frac{f(p)}{2} \cdot \int_{p-\varepsilon}^{p+\varepsilon} dx$$

(2) $$= \frac{f(p)}{2}\left(p+\varepsilon - (p-\varepsilon)\right) = \varepsilon \cdot f(p)$$

(we could use FTC, and integral properties because f cont.s $\Rightarrow f$ is Riemann integrable)

Combining (1) and (2),

$$\int_a^b f(x)\,dx \geq 0 + \varepsilon \cdot f(p) + 0 \quad \text{for some particular } \varepsilon > 0$$
$$\neq 0$$

This contradicts our given statement that $\int_a^b f\,dx = 0$

$\Rightarrow$ there cannot exist $x = p$ s.t. $f(p) \neq 0$

$\Rightarrow f(x) = 0$ on $[a,b]$ $\square$

[Note: f is not necessarily 0 on $\mathbb{R}$.
Take $f = \begin{cases} 0 & x \in [a,b] \\ 1 & x \notin [a,b] \end{cases}$ as example]

d. Take any partition $P = \{a = t_0 < t_1 < \ldots < t_n = b\}$

(1) So, $U(f,P) = \sum_{k=1}^{n} M(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) = \sum_{k=1}^{n} 1 \cdot (t_k - t_{k-1}) = b - a$

$\because$ by the density principle, for any $t_{k-1}, t_k \in \mathbb{R}$, $\exists\, q \in \mathbb{Q}$ s.t. $t_{k-1} < q < t_k$ and $f(q) = 1$

(2) Also, $L(f,P) = \sum_{k=1}^{n} m(f, [t_{k-1}, t_k]) \cdot (t_k - t_{k-1}) = 0 \cdot (t_k - t_{k-1}) = 0$

$\because$ by density, for any $t_{k-1}, t_k \in \mathbb{R}$, $t_{k-1} = \sqrt{2} \cdot a_{k-1}$, $t_k = \sqrt{2} \cdot a_k$ for some $a_{k-1}, a_k \in \mathbb{R}$, and $\exists\, r \in \mathbb{Q}$ s.t. $a_{k-1} < r < a_k$

$\Rightarrow t_{k-1} = \sqrt{2} \cdot a_{k-1} < \sqrt{2} \cdot r < \sqrt{2}\, a_k = t_k$ where $\sqrt{2} \cdot r \notin \mathbb{Q} \Rightarrow f(\sqrt{2} \cdot r) = 1$

Combining (1) and (2), $U(F,P) = b - a \neq 0 = L(F,P)$
(since $a \neq b$)

$\Rightarrow f$ is not Riemann-integrable $\square$

⌜Note that this proof is only an elaboration of the proof on page 272 of Ross⌟

2a. False, take as a counter-example:

$$f(x) = \begin{cases} -1 & x \in \mathbb{Q} \\ 1 & x \in \mathbb{Q} \end{cases}, \text{ bdd and real}$$

Note, $f^2 \equiv 1 \quad \forall x$, which is cont.s and therefore Riemann-integrable.

But, $f(x)$ is not Riemann integrable:

$$U(F,\alpha) = \sum_{k=0}^{n-1} M(f,[\alpha(k), \alpha(k+1)]) \cdot (\alpha(k+1) - \alpha(k))$$
$$= 1 \cdot (b-a) = b - a$$

since from (1d) there is a rational in any interval

$$L(F,\alpha) = \sum_{k=0}^{n-1} m(f,[\alpha(k), \alpha(k+1)]) \cdot (\alpha(k+1) - \alpha(k)) = -1(b-a) = a-b$$

since from (1d), there is an irrational in any interval

$\Rightarrow U(F,\alpha) \neq L(F,\alpha)$ since $a \neq b$

$\Rightarrow f$ is not Riemann-integrable $\square$